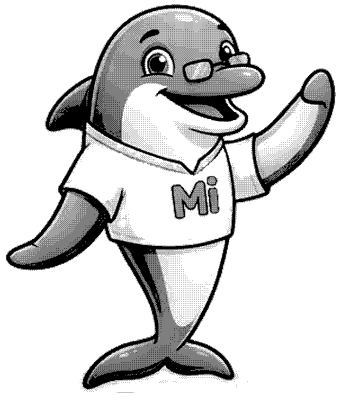
Math Inquirers

Extension Workbook

with answers

Level 6

Self-study | Ages 11-13



My name _____

About this book

Math Inquirers Extension Workbook, Level 6
Self-study practice with answers.

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Good to know

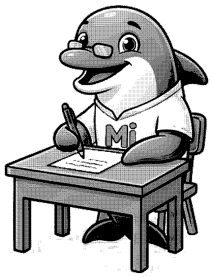
- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

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Welcome!

This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- The rulers in Unit 9 are printed at real size.

The four page names

- **Start here.** The main path. Do these first.
- **Think harder.** Longer problems. Hints are printed upside down.
- **Go further.** A weekend page. Big challenges, just for fun.
- **Check yourself.** Four quick questions at the end.

Contents

	About this book Copyright and licence	page 2
	Welcome	page 3
	My pace plan	page 5
1	**Code the pattern** Not in this proof edition	–
2	**Prime hunters** Not in this proof edition	–
3	**Below zero** Not in this proof edition	–
4	**Algebra machines** Not in this proof edition	–
5	**Every possible way** Not in this proof edition	–
6	**Fair or unfair?** Not in this proof edition	–
7	**Turtle geometry** Not in this proof edition	–
8	**Secret codes** Not in this proof edition	–
9	**Data detectives** Data detectives · follows Holiday Program Lesson 9 · answers on page 16	page 6
10	**Showcase challenge** Not in this proof edition	–
	My progress	page 13
	My certificate	page 14
	Answers Check your work, with short reasons	page 15

My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Code the pattern	Not in this proof edition				☐
2	Prime hunters	Not in this proof edition				☐
3	Below zero	Not in this proof edition				☐
4	Algebra machines	Not in this proof edition				☐
5	Every possible way	Not in this proof edition				☐
6	Fair or unfair?	Not in this proof edition				☐
7	Turtle geometry	Not in this proof edition				☐
8	Secret codes	Not in this proof edition				☐
9	Data detectives	6-7	8-9	10 and 12	11	☐
10	Showcase challenge	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

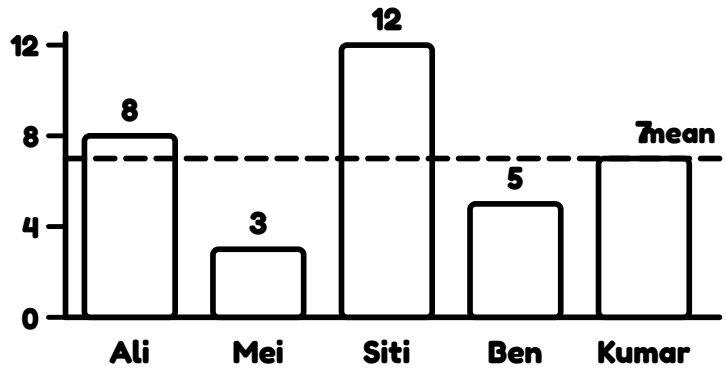
Learn it: Mean, median and mode

My goal. I can find the mean, a missing value from the mean, and compare the mean with the median and mode.

The mean is the total divided by how many values there are. It is the level you get when the towers are made equal. Total = mean × how many. One very large or very small value can pull the mean away from a typical value.

Worked example

Five friends collect shells: Ali 8, Mei 3, Siti 12, Ben 5 and Kumar 7. Find the mean number of shells.



1. **Add all the values: 8 + 3 + 12 + 5 + 7 = 35.**
2. **Divide by how many values there are: 35 ÷ 5 = 7.**
3. **Check by levelling. The towers above 7 have 1 + 5 = 6 extra shells. The towers below 7 are 4 + 2 = 6 short.**

Answer: The mean is 7 shells.

How to work with a mean

1. Add all the values. A value of 0 still counts.
2. Divide the total by how many values there are.
3. Missing value: total = mean × how many, then take away the values you know.
4. Median is the middle value in order. Mode is the most common value.

Your turn

Answer at the back of the book.

Try **The mean of 4 numbers is 12. Three of them are 9, 15 and 10. The 4th number is** ☐.

Name _____

Start here

Do these first. Take your time.

1 **Find the mean of each list.**

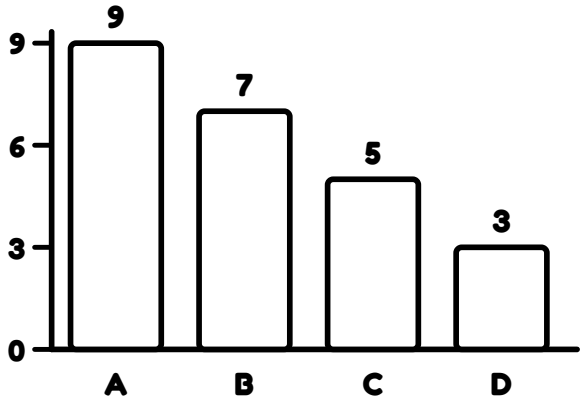
(a) 14, 9, 11, 6: mean = ☐

(b) 9, 0, 6, 0, 10: mean = ☐

(c) 1.5, 2.5, 4, 3: mean = ☐

Working: _____

2 **Move blocks from tall towers to short towers until all towers are equal.**



(a) The mean height is ☐ blocks.

(b) ☐ blocks must move from the tall towers to the short towers.

3 **The mean mass of 5 parcels is 6 kg.**

(a) The 5 parcels weigh ☐ kg in all.

(b) 3 of them weigh 4 kg, 9 kg and 7 kg. The other 2 weigh ☐ kg together.

4 **4 test scores have a mean of 82. Three of the scores are 78, 90 and 85.**

(a) The 4th score is ☐.

(b) A 5th score of ☐ keeps the mean at 82.

Start here

Keep going. Check your work.

5 **The mean of 10, 12 and 14 is 12. Predict, then find what happens when the list changes.**

(a) Add the value 12. The mean goes up stays the same goes down

(b) Add the value 20. The new mean is ☐.

(c) Take away the value 14. The new mean is ☐.

6 **The median is the middle value when the values are in order. The mode is the value that appears most often.**

(a) For 12, 7, 9, 7, 5 the median is ☐. **(b)** The mode is ☐.

(c) The mean is ☐. **(d)** For 2, 8, 4, 6 the median is ☐.

7 **A cafe has 7 staff. Their pay in dollars a month is 1500, 1600, 1600, 1700, 1800, 1900 and 9500.**

(a) The mean pay is ☐ dollars.

(b) The median pay is ☐ dollars.

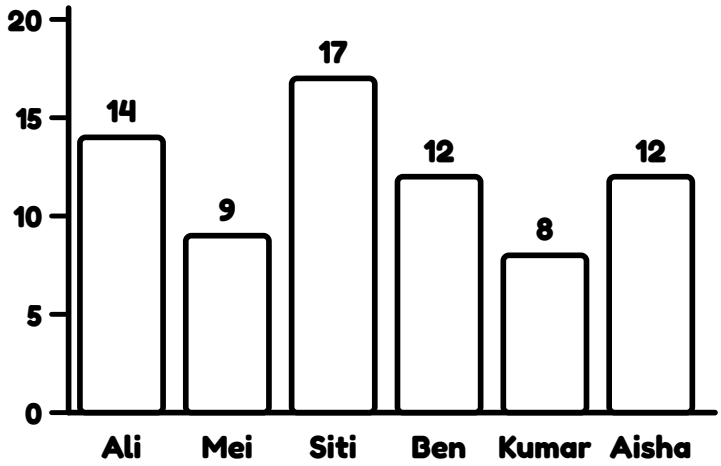
(c) Which describes what most staff earn better? mean median

8 **The bar chart shows the number of stickers 6 pupils have.**

(a) They have ☐ stickers in all.

(b) The mean is ☐ stickers.

(c) ☐ pupils have more than the mean.



Start here

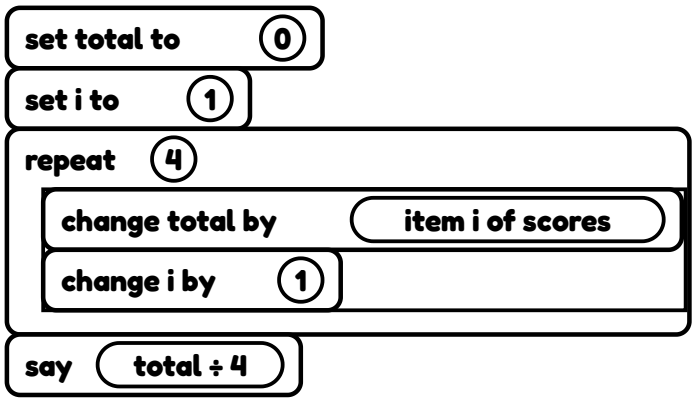
Nearly there. You can do it.

9 **The list scores holds 7, 9, 3 and 5. A loop adds up the items and the program says the mean.**

(a) After the loop, total is ☐.

(b) The program says ☐.

(c) If the last line says total ÷ 3, it says ☐.



10 **The mean of 4, 8 and 12 is 8. Change every number in the list in the same way.**

(a) Add 5 to every number. The new mean is ☐.

(b) Double every number. The new mean is ☐.

11 **The mean mass of 5 children is 40 kg. A 6th child of mass 52 kg joins them.**

(a) The 5 children weigh ☐ kg in all.

(b) With the 6th child the total is ☐ kg.

(c) The new mean is ☐ kg.

Think harder

Hints are upside down. Try first!

1 **Five whole numbers have a mean of 7 and a median of 6. Their only mode is 4, and the largest number is 13.**

The numbers are _____

Working: _____

_____

_____

Hint. The 2 smallest are 4 and 4. Use the total for the other.

2 **A class has 12 girls and 18 boys. The girls have a mean mass of 40 kg. The boys have a mean mass of 45 kg.**

(a) The girls weigh ☐ kg in all. **(b)** The boys weigh ☐ kg in all.

(c) The mean mass of all 30 is ☐ kg.

(d) It is more than 42.5 because there are more girls boys

Working: _____

_____

Hint. Find each group total first. Then find the total for all 30.

3 **The mean of 3 numbers is 6.4. Two of them are 5.5 and 7.2.**

(a) The 3 numbers add up to ☐.

(b) The third number is ☐.

Working: _____

_____

_____

Hint. Total = mean × how many. Then take away what you know.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 6: Pairing up.**
The numbers 7, 12, 17, 22, ..., 102 go up by 5 each time. There are 20 of them.

(a) The 1st and the last add up to ☐.

(b) The 20 numbers make ☐ such pairs.

(c) The mean of the 20 numbers is ☐.

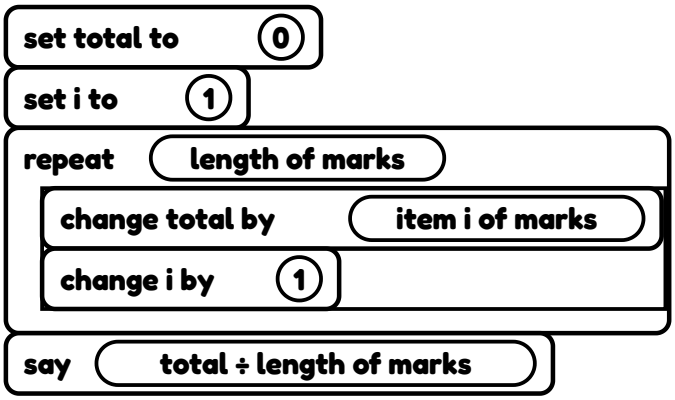
Hint. Pair the 1st with the last, the 2nd with the 2nd last.

2 **The list marks holds 4, 8, 6, 10 and 7. This program adds up the items and says the mean.**

(a) It says ☐.

(b) With repeat 4 and ÷ 5 it says ☐.

(c) With 13 added to the list it says ☐.



Hint. Add up the items the loop reaches. Then divide.

3 **Beyond Level 6: A frequency table.**
In 10 matches a team scored goals as shown.

Goals	0	1	2	3
Matches	2	4	3	1

(a) The team scored ☐ goals in all.

(b) The mean number of goals is ☐.

(c) The median number of goals is ☐.

Hint. 2 matches with 0 goals, 4 with 1 goal: count goals for each column.

✔ Check yourself

No hints. Try each one on your own.

1 **The mean of 6, 10, 2 and 14 is ☐.**

Working: _____

2 **For 4, 9, 4, 6, 11 the median is ☐ and the mode is ☐.**

Working: _____

3 **The mean of 3 numbers is 14. Two of them are 10 and 18. The third is ☐.**

Working: _____

4 **The mean of 4 tests is 70. A 5th test score is 80. The new mean is ☐.**

Working: _____

My score: ☐ out of 4. Check the answers at the back.

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Code the pattern	☐	☐	☐	☐	/ 4	
2	Prime hunters	☐	☐	☐	☐	/ 4	
3	Below zero	☐	☐	☐	☐	/ 4	
4	Algebra machines	☐	☐	☐	☐	/ 4	
5	Every possible way	☐	☐	☐	☐	/ 4	
6	Fair or unfair?	☐	☐	☐	☐	/ 4	
7	Turtle geometry	☐	☐	☐	☐	/ 4	
8	Secret codes	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Showcase challenge	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 6.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 9 · Data detectives

Your turn
page 6

Answer: 14
Why: The total of 4 numbers is 4 × 12 = 48. The 3 known numbers add to 9 + 15 + 10 = 34. The 4th number is 48 − 34 = 14.

START HERE

1
page 7

Answer: (a) 10 (b) 5 (c) 2.75
Why: (a) 14 + 9 + 11 + 6 = 40, and 40 ÷ 4 = 10. (b) 9 + 0 + 6 + 0 + 10 = 25, and 25 ÷ 5 = 5 (the 0s count as values). (c) 1.5 + 2.5 + 4 + 3 = 11, and 11 ÷ 4 = 2.75.

2
page 7

Answer: (a) 6 (b) 4
Why: Total: 9 + 7 + 5 + 3 = 24 blocks, shared over 4 towers: 24 ÷ 4 = 6. Tower A gives away 3 blocks and tower B gives away 1: 4 blocks move. Towers C and D receive them (1 and 3).

3
page 7

Answer: (a) 30 (b) 10
Why: Total mass = mean × how many = 6 × 5 = 30 kg. The 3 known parcels weigh 4 + 9 + 7 = 20 kg, so the other 2 weigh 30 − 20 = 10 kg together.

4
page 7

Answer: (a) 75 (b) 82
Why: Total of 4 scores = 4 × 82 = 328. The known scores add to 78 + 90 + 85 = 253. The 4th score is 328 − 253 = 75. A 5th score equal to the mean, 82, keeps the mean at 82.

5
page 8

Answer: (a) stays the same (b) 14 (c) 11
Why: (a) A new value equal to the mean does not change it: the mean stays at 12. (b) 10 + 12 + 14 + 20 = 56, and 56 ÷ 4 = 14. (c) 10 + 12 = 22, and 22 ÷ 2 = 11.

6
page 8

Answer: (a) 7 (b) 7 (c) 8 (d) 5
Why: 12, 7, 9, 7, 5 in order: 5, 7, 7, 9, 12. The middle one is 7, the mode is 7 and the mean is 40 ÷ 5 = 8. 2, 8, 4, 6 in order: 2, 4, 6, 8. The 2 middle values are 4 and 6, so the median is halfway between them: 5.

7
page 8

Answer: (a) 2800 (b) 1700 (c) median
Why: Total: 1500 + 1600 + 1600 + 1700 + 1800 + 1900 + 9500 = 19600, and 19600 ÷ 7 = 2800. In order the middle (4th) value is 1700. Only the owner earns 9500. The mean is pulled up by that one value, so the median 1700 is more typical.

8
page 8

Answer: (a) 72 (b) 12 (c) 2
Why: Total: 14 + 9 + 17 + 12 + 8 + 12 = 72. Mean = 72 ÷ 6 = 12. Above 12: 14 and 17 is 2 pupils. (Two pupils have exactly 12.)

9
page 9

Answer: (a) 24 (b) 6 (c) 8
Why: i goes 1, 2, 3, 4 and the loop adds 7, 9, 3 and 5: total 24. The last line says 24 ÷ 4 = 6. If it divided by 3 it would say 24 ÷ 3 = 8.

10 page 9
Answer: (a) 13 (b) 16
Why: Add 5 to each: 9, 13, 17. 39 ÷ 3 = 13, which is the old mean + 5. Double each: 8, 16, 24. 48 ÷ 3 = 16, which is the old mean × 2.

11 page 9
Answer: (a) 200 (b) 252 (c) 42
Why: Total of 5 children = 5 × 40 = 200 kg. With the 6th child: 200 + 52 = 252 kg. New mean = 252 ÷ 6 = 42 kg.

THINK HARDER

1 page 10
Answer: 4, 4, 6, 8, 13
Why: The total is 5 × 7 = 35. In order the middle number is 6. 4 is the mode and the smallest, so the first two numbers are 4 and 4: 4, 4, 6, ?, 13. 4 + 4 + 6 + 13 = 27, so the 4th number is 35 − 27 = 8. The numbers are 4, 4, 6, 8, 13.

2 page 10
Answer: (a) 480 (b) 810 (c) 43 (d) boys
Why: Girls: 12 × 40 = 480 kg. Boys: 18 × 45 = 810 kg. All 30: 480 + 810 = 1290 kg. 1290 ÷ 30 = 43. The mean of the whole class is 43 kg, not 42.5, because there are more boys.

3 page 10
Answer: (a) 19.2 (b) 6.5
Why: The total is 3 × 6.4 = 19.2. The known numbers add to 5.5 + 7.2 = 12.7. The third number is 19.2 − 12.7 = 6.5.

GO FURTHER

1 page 11
Answer: (a) 109 (b) 10 (c) 54.5
Why: The 1st and last add to 7 + 102 = 109. So do the 2nd and the 2nd last (12 + 97). 20 numbers make 10 pairs of 109, so the total is 1090. The mean is 1090 ÷ 20 = 54.5, which is halfway between 7 and 102.

2 page 11
Answer: (a) 7 (b) 5.6 (c) 8
Why: Total = 4 + 8 + 6 + 10 + 7 = 35, and 35 ÷ 5 = 7. With repeat 4 the loop adds only 4 + 8 + 6 + 10 = 28, and 28 ÷ 5 = 5.6. A 6th item 13 gives a total of 48, and 48 ÷ 6 = 8 (the program must divide by 6, the length of the list).

3 page 11
Answer: (a) 13 (b) 1.3 (c) 1
Why: Total goals = 0 × 2 + 1 × 4 + 2 × 3 + 3 × 1 = 0 + 4 + 6 + 3 = 13. Mean = 13 ÷ 10 = 1.3. In order the 10 values are 0, 0, 1, 1, 1, 1, 2, 2, 2, 3. The 5th and 6th values are 1 and 1, so the median is 1.

CHECK YOURSELF

1 page 12
Answer: 8
Why: 6 + 10 + 2 + 14 = 32, and 32 ÷ 4 = 8.

2 page 12
Answer: 6, 4
Why: In order: 4, 4, 6, 9, 11. The middle value is 6, and 4 appears most often.

3 page 12
Answer: 14
Why: Total = 3 × 14 = 42. 42 − 10 − 18 = 14.

4 page 12

Answer: 72
Why: The 4 tests total 4 × 70 = 280. With the 5th: 280 + 80 = 360, and 360 ÷ 5 = 72.